

Daeki Hong

Senior Unity/XR Systems Engineer | AI Product Builder

10 years of Unity/XR systems work across industrial XR, BIM runtime tooling, digital-twin authoring, robotics interfaces, and applied AI systems. My current center is DeskRelay: a self-host AI development control surface created through full-vibe coding with Claude Code. I use AI as a development organization, not a shortcut: role boundaries, instructions, state boards, recovery loops, UX review, and cold validation.

DeskRelay**AI as Work System****Failure-State Validation****Unity C# Runtime****Industrial XR****BIM / Robotics**

CONTACT

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BEST FIT

Self-host AI product platforms, local agent runtimes, connector tooling, Unity-based simulation, robotics interfaces, BIM / digital-twin tools, and delivery-critical industrial software.

EDUCATION

Korea Aerospace University, B.S. in Software.

Positioning: AI product work first, with DeskRelay as the flagship. The value is not passive AI delegation; it is directing AI output through product judgment, operational UX, failure-state testing, and scope discipline. Unity industrial development is the execution base behind that judgment.

CORE PROOF

AI PRODUCT / FULL-VIBE CODING

DeskRelay

Self-host browser control for Claude Code running on local PCs. Built through Claude Code full-vibe coding while I directed product intent, manager/worker roles, instructions, state, recovery, UX language, and validation.

SERVER / NETWORK

Connector Runtime

Site frontend/backend, connector daemons, device registration, LAN/Tailscale operation, WebSocket/SSE relays, bearer-token authorization, session queues, updates, diagnostics, and offline failure reporting.

UNITY INDUSTRIAL

WatchBIM / Unity-CrossSection

Heavy BIM geometry interaction, mesh-plane cross sections, 2D analysis, Burst/Job System processing, NativeArray handling, mesh caching, shader-based visualization, and camera tooling.

ROBOTICS / XR

VR Robot / MetaArm

Quest 3 input to Unity runtime logic, coordinate transforms, smoothing, Flask APIs, TCP communication, RoboDK, safety constraints, calibration, and public demo readiness.

AI product direction on top of industrial Unity delivery

DeskRelay - Independent AI Product / Systems Track

2025 - PRESENT

Self-host AI development control surface, local agent runtime, connector platform.

- Created DeskRelay through full-vibe coding with Claude Code, using AI as an execution partner while I drove product direction, role boundaries, instruction management, state boards, review loops, and operational acceptance.
- Challenged happy-path output through registration, offline states, update visibility, permissions, LAN/Tailscale boundaries, mobile usability, diagnostics, and whether each UI state exposed only actionable information.
- Refined generated UI copy, README/manual text, privacy wording, and status messages; scoped the product toward a self-host open-source tool focused on installation trust, connector reliability, and local control.

YMX - Senior Engineer, Client Team 2

DEC 2024 - PRESENT

Unity industrial development: digital-twin authoring, robotics interfaces, exhibition content.

- Built DXCenter, a Unity 6 URP/WebGL authoring environment with EditorWindow tools, ScriptableObject data flow, Service Locator, EventBus, Strategy-based state effects, adapter-style device integration, and non-developer runtime preparation.
- Built Unity - Flask - RoboDK robot control flows for Quest 3 based teleoperation, including coordinate conversion, smoothing, speed/acceleration limits, TCP/HTTP communication, calibration, and safety handling.

Sixdopamine - Project-based

2025

Applied AI experience system: Flask AI pipeline, XR runtime, operations dashboard.

- Connected Whisper, GPT, TTS, Skybox AI, Meshy, GCP storage, XR timing, and dashboard visibility into the Neostalgia exhibition pipeline.
- Recognition: BIFAN XR Beyond Reality Official Selection and MSIT Minister Citation.

Jangheon Industrial - Seoul Office

JUN 2020 - DEC 2024

BIM visualization, geometry tooling, drawing automation, engineering software.

- Built WatchBIM features for highway BIM runtime data: custom hit engine, mesh optimization, auto LOD, cross-section generation, 2D analysis viewer, slope labeling, minimap generation, and camera tools.
- Implemented Unity-CrossSection as a reusable C# module using coroutines, NativeArray, Burst-compatible jobs, progress/cancel flow, and mesh-plane intersection processing.
- Built WPF road-design drawing automation using MVVM, MySQL, and CAD entity generation.

Mline Studio / Saltworks / Swink

2015 - 2019

Unity gameplay, VR simulators, hardware-linked XR delivery, educational software.

- Delivered FPS safety training, construction/factory/naval VR simulators, AR safety education, RS232-linked devices, multiplayer mission flow, installation, live operation, teardown, and operator handoff.